

[illegible]

Accel & Gyro	
CS_ADXL	CS_ADXL
CS_ICM	CS_ICM
ADXL_INT0	INT0_ADXL
ADXL_INT1	INT1_ADXL
ICM_INT1	INT1_ICM
SCLK	SCL_Gyro_Accel
MOSI	SDI_Gyro_Accel
MISO	SDO_Gyro_Accel
FSYNC_ADXL	FSYNC_ADXL
FSYNC_ICM	FSYNC_ICM

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Magnetmeter and Barometer

CS_BARO	→ Baro_CS
Baro_INT	→ Baro_INT
SCLK	→ Mag_Barom_SCK
MOSI	→ Mag_Barom_SDI
MISO	→ Mag_Barom_SDO
CS_MAG	→ Mag_CS
Mag_INT	→ Mag_INT

File: Mag_Baro.kicad_sch

J1	J1-J2 STM	J2	
Conn_01x17		Conn_01x17	
17	B9	17	B10
16	B8	16	B2
15	B7	15	B1
14	B6	14	B0
13	B5	13	MOSI
12	FSYNC_ICM	12	MISO
11	FSYNC_ADXL	11	SCLK
10	ICM_INT1	10	CS_ICM
9	ADXL_INT1	9	CS_ADXL
8	ADXL_INT0	8	CS_BARO
7	A10	7	CS_MAG
6	A9	6	A0
5	A8	5	R
4	B15	4	C15
3	Mag_INT	3	C14
2	Baro_INT	2	C13
1	B12	1	VB

J4
Conn_01x04

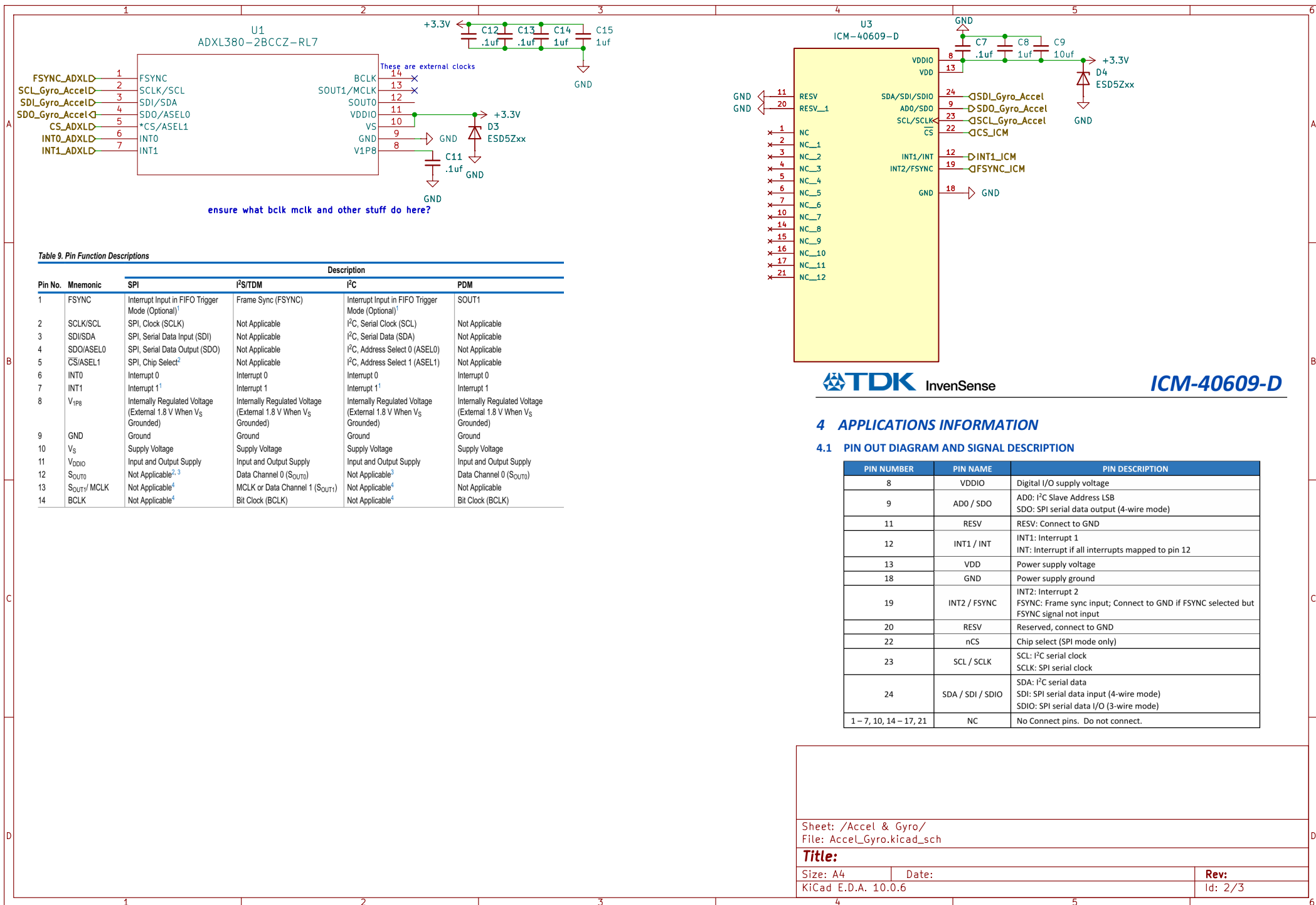
Pin	Signal
4	CS_BARO
3	CS_MAG
2	Baro_INT
1	Mag_INT

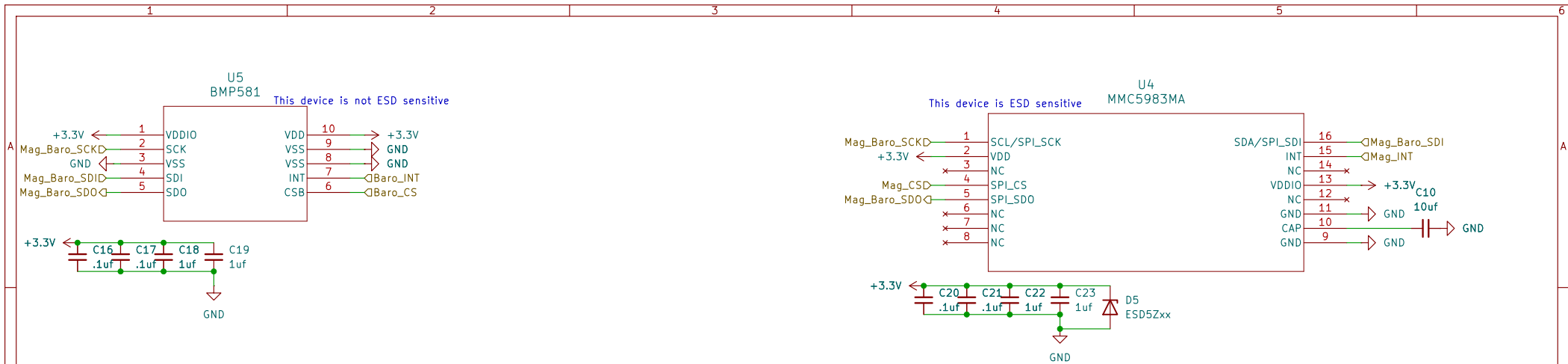
J3 Accel/Gyro J3

Conn_02x06_Top_Bottom

Signal	Pin 1	Pin 2	Pin 3	Pin 4	Pin 5	Pin 6	Pin 7	Pin 8	Pin 9	Pin 10	Pin 11	Pin 12
+5V							1	8				CS_ADXL
+3.3V							2	7				CS_ICM
MISO							3	9				ADXL_INT0
MOSI							4	10				ADXL_INT1
SCLK							5	11				FSYNC_ADXL
ICM_INT1							6	12				FSYNC_ICM

Rev:
Id: 1/3





6.2.2 SPI 4-wire

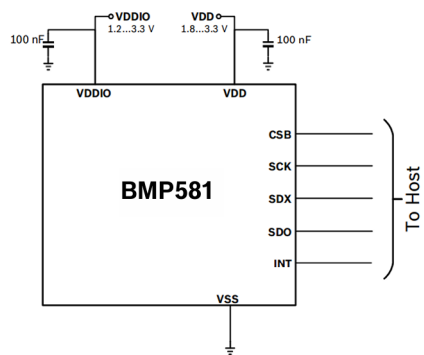
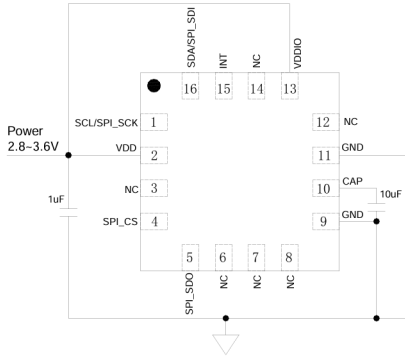


Figure 25: SPI 4-wire connection diagram

Pin	Name	I/O Type	Description	Connect to	
				SPI 4W	SPI 3W
1	VDDIO	Supply	Digital interface supply	VDDIO	
2	SCK	In	Serial clock input	SCK	SCK
3	VSS	Supply	Ground	GND	
4	SDI	In/Out	Serial data input	SDI	SDI/SDO
5	SDO	In/Out	Serial data output	SDO	DNC
6	CSB	In	Chip select	CSB	CSB
7	INT	Out	INT output	host INT input, GND ^a or DNC	
8	VSS	Supply	Ground	GND	
9	VSS	Supply	Ground	GND	
10	VDD	Supply	Analog supply	VDD	

a. GND connection is allowed, as long as the IRQ pin is not activated



<Top View>
SPI Single Supply Connection

PIN DESCRIPTIONS

CAP– This is the supply input for SET/RESET coil. MEMSIC recommends a minimum 10 uF low ESR capacitor placed in close proximity to the CAP pin.

GND– This is the ground pin for the magnetic sensor.

SDA/SPI_SDI – This pin is the I²C/SPI serial data line, and operates in I²C FAST (400 KHz)/SPI mode.

SCL/SPI_SCK– This pin is the I²C/SPI serial clock line, and operates in FAST (400 KHz)/SPI mode.

SPI_CS – This pin is chip select line for SPI (active low). Tie to VDDIO for I²C Interface.

SPI_SDO – This pin the SPI serial data out line.

VDD–This pin is the power supply of the ASIC.

VDDIO–This pin is the power supply of the digital interface of the ASIC.

INT– This pin is Interrupt output. Active High. It is held as hi-Z until an interrupt function is enabled.

Note: VDD and CAP pins are shorted together inside the device.

Sheet: /Magnetmeter and Barometer/
File: Mag_Baro.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.6

Date:

Rev:

Id: 3/3